siRNA-GPT：PPT 图册与结论说明

五张精简正文图 + 三项真实备份素材 + 两张原版记录图 | 图内英文，解释用中文

建议汇报顺序

01 **既购 si-VEGF 的真实 Agent 案例**
系统按计划协调工具，汇总可追溯证据。

02 **文献问题、原文与实际回答**
回答确实使用检索到的来源；仅展示这一次问答。

03 **完整 65 条候选的计算结果**
展示候选来源、模型分数和种子匹配分布。

04 **完整 264 项近全长查询**
展示全量检查；23/65 条存在非靶近似匹配。

05 **自有 RNA-seq 的 VEGFA 基线**
同批次缺氧组均值更高；仅作表达背景。

06–08 **真实网页、检索界面与旧版案例**
备份使用，保留各自的版本与限制。

09–10 **原版完整案例图**
保留全部细节，方便追溯与答辩追问。

当前证据支持“系统已完成真实案例与可追溯计算整理”。独立质量评分和 siRNA 实测效力仍未完成，不在本图册中补写。

seed：种子区短序列匹配；TPM：转录本相对表达量单位。seed计数与近全长匹配数不能直接比较。

图 01 | 既购 si-VEGF：真实 Agent 运行

正文优先 | 精简重排；不是网页截图；不是新模型运行

One purchased si-VEGF: a recorded Agent run

Saved records, re-typeset | Chinese request translated | run 20261024_121627_1e50c9

Request Review this duplex using its bound results and expression context.

a Six confirmed steps, all executed

01	Check candidate	executed
02	Match VEGFA reference	executed
03	Read OligoFormer result	imported
04	Read seed result	imported
05	Join expression data	executed
06	Save report	executed

8 model requests = 1 planning + 7 execution; 6 tools.

b Verified tool fields

19-nt paired cores (5'-3'; conditional roles)

UUUCUUGUCUUGCUCUAUC

GAUAGAGCAAGACAAGAAA

1.1219 model score, not %

2,839 non-target seed genes (guide)

18/20 fixed reference transcripts

Core-only analysis; TT/N3 not modeled.

OligoFormer and seed results were imported from verified local computations.

Raw answer retained: one 19-nt core was miscopied as 24 nt. This table uses tool evidence.

这是什么

一条既购双链的真实运行，按确认计划完成六步核查。

要说明什么

Agent 能否把序列、已有计算结果和表达背景接起来?

得到什么结论

流程跑通并保存报告；8次模型请求对应1个案例。

边界：分数和 seed 为本地结果导入；原答有19位错写24位，不能称回答全对。

图 02 | 文献证据支持的真实回答

正文优先 | 原问答译文摘录；论文、源码与项目说明分开

An answer supported by retrieved evidence

B02 saved records | 1 model call | 6 retrieved excerpts | Translated question and answer excerpts

Does a score above 1 mean more than 100% knockdown?

a Evidence actually passed to the model

[S3] OligoFormer article

"prediction score for each siRNA"

DOI: 10.1093/bioinformatics/btae577

[S2, S5] Official inference source

Y_PRED = [i*1.341 for i in Y_PRED]

scripts/infer.py @ e2f53ad63387

b Saved answer

No.

Prediction score

is not measured knockdown. [S1, S3]

Scaled by 1.341. [S2, S5]

English translations are labeled; source quotations are retained. This is not a webpage screenshot.

S1 is a project setting. This single question does not establish overall accuracy or Agent superiority.

这是什么

B02 的真实问题、实际传入的原文/源码和回答摘录。

要说明什么

模型是否真正取得来源，并按来源解释专业概念?

得到什么结论

这一次问答正确区分模型分数与实测敲低率，引用可核对。

边界：单题演示不代表总体准确率；项目说明不是外部论文结果。

图 03 | 候选来源、完整分数与 seed 注释

正文可用 | 原104条、修订65条和历史9条的关系已标明

A complete, traceable candidate pool

All 65 design predictions shown | Purchased si-VEGF is a separate entry | Existing calculations re-plotted

a Candidate accounting

Original designs

104

Reporting pool

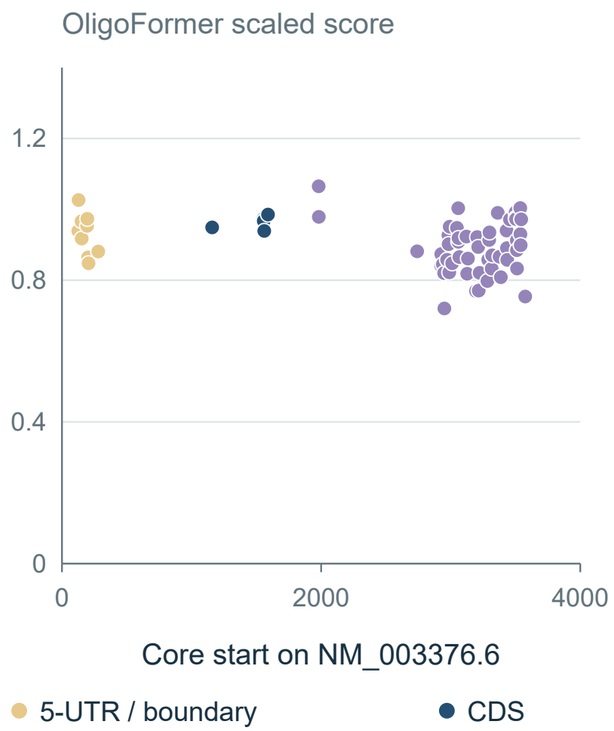
65

Historical subset

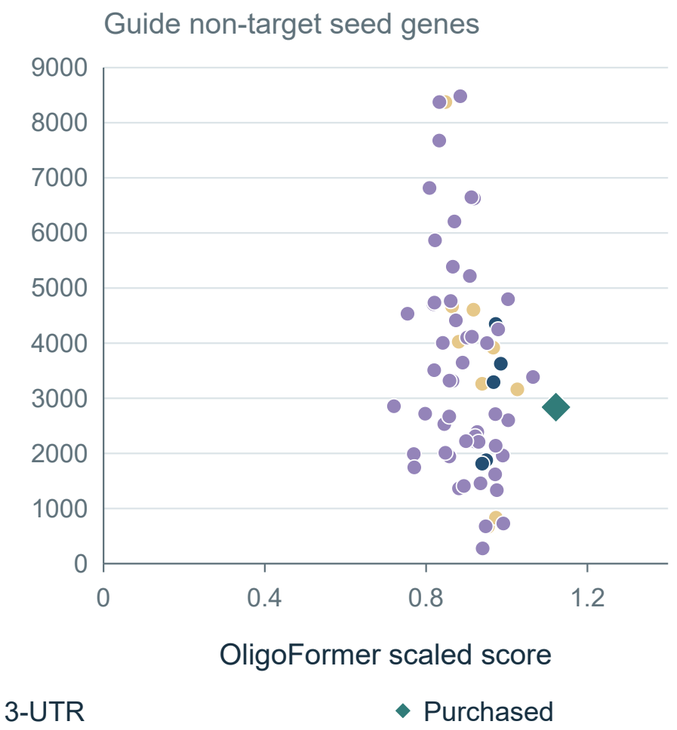
9

Nested counts; not final selections.

b Score by target position



c Score and seed annotation



Scores use the official 1.341 scale; they are not measured knockdown percentages.

Seed matches are sequence annotations. All sequence flags remain in the source table; no final wet-lab shortlist.

这是什么

全部65条设计的模型/seed结果，既购序列单列。

要说明什么

候选有无来源？是否只挑了少数高分结果展示？

得到什么结论

完整池均有计算结果；各候选的分数与匹配注释存在差异。

边界：9条是历史子集；高分不等于高实测效力；seed匹配不等于实验脱靶。

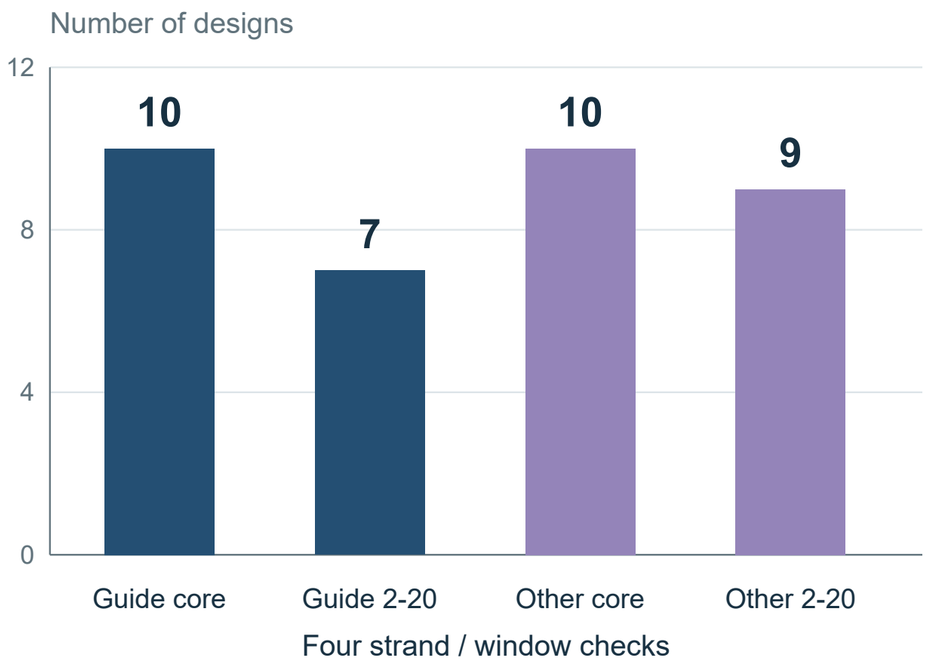
图 04 | 264 项近全长检查已全覆盖

正文可用 | 65条设计+既购1条；每条按四种查询检查

Near-full checks: all 264 queries completed

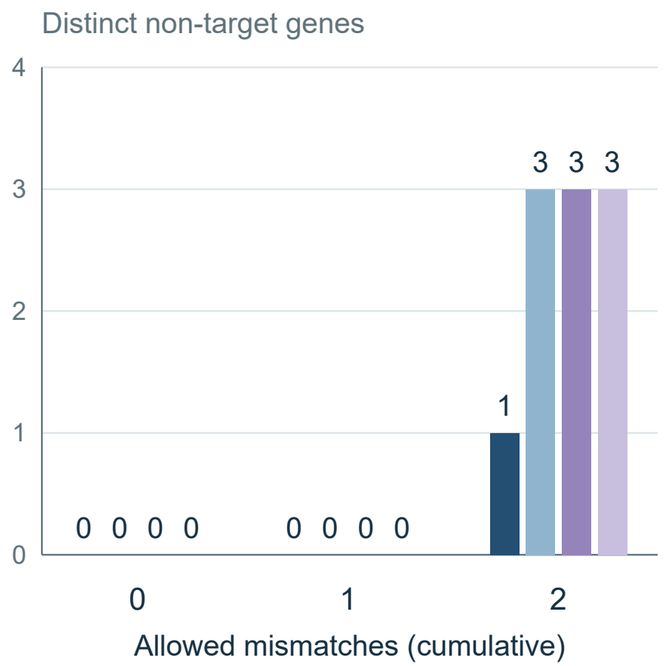
65 designs + one purchased duplex | Four overlapping 19-nt queries per entry | Fixed 186,185-transcript reference

a Designs with a non-target hit at <=1 mismatch



Union across the four checks: 23 / 65 designs (overlap allowed).

b Purchased duplex: non-target genes



Bar order: guide core, guide 2-20, other core, other 2-20.

These overlapping queries are correlated checks, not four independent biological experiments.

Sequence matches are not measured off-target effects. The other 42 designs are not certified safe.

这是什么

两条链各按1–19、2–20位比对，共264项查询。

要说明什么

所有候选是否都检查？是否有接近全长的非靶匹配?

得到什么结论

264项查询完成；65条中23条有0/1错配非靶匹配。

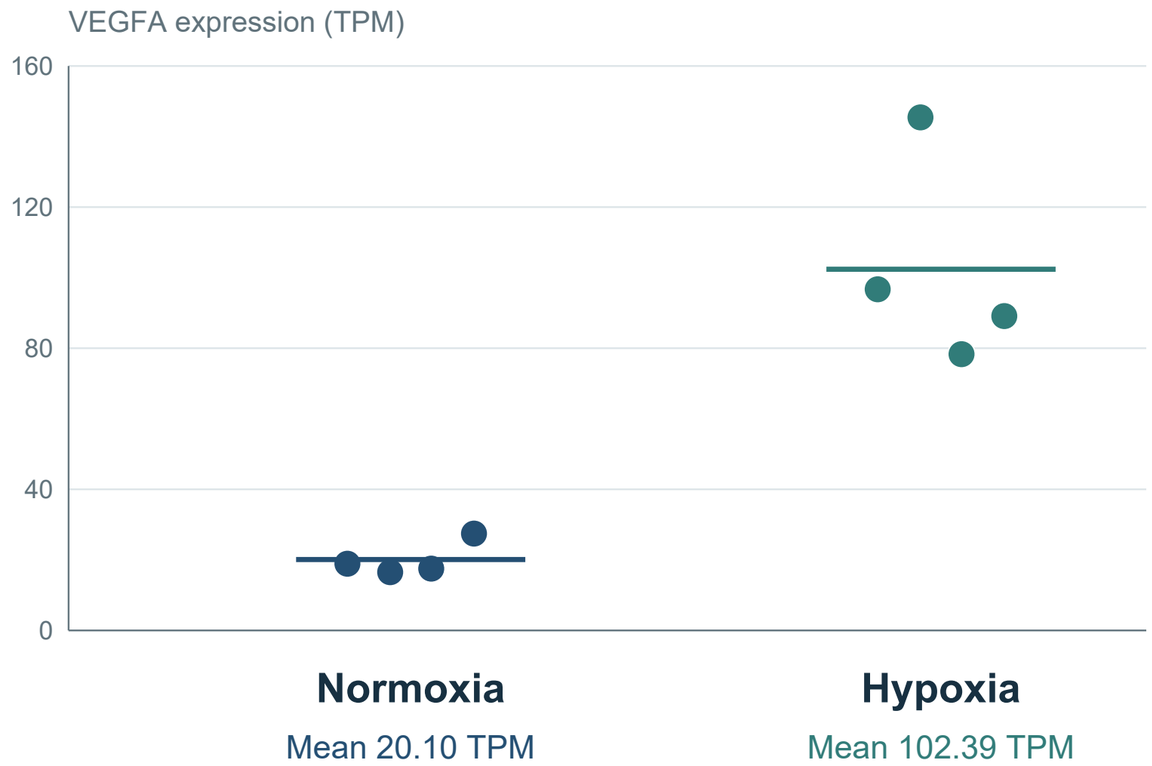
边界：10、7、10、9不可相加成独立候选数；其余42条不能据此称安全。

图 05｜自有 RNA-seq：VEGFA 表达基线

正文可用｜原始8个TPM点重绘；不增加统计显著性

VEGFA expression in untreated ARPE-19 dishes

Same culture batch | Four dishes per condition | Hypoxia: 1% O2 for 24 h | No siRNA or transfection reagent



5.09x

ratio of group means

Descriptive baseline

for this culture batch.

Dots are the eight actual dish-level TPM values; horizontal lines show group means.

This is expression context, not siRNA knockdown, four independent culture rounds, or a treatment effect.

这是什么

同批ARPE-19常氧/缺氧各4皿，均未转染。

要说明什么

自己的RNA-seq能为课题提供什么直接背景证据?

得到什么结论

该批次VEGFA均值为20.10与102.39 TPM，比值约5.09。

边界：仅描述这批样本；不是4轮独立培养、siRNA敲低效果或状态风险重排序。

图 06｜原网站的真实完成界面

备份素材｜2026-10-24既购案例；原图原样放入



这是什么

真实网站完成既购序列核查后的截图。

要说明什么

系统是否具有实际可操作的网页入口与结果界面？

得到什么结论

该案例确实在原网站完成；界面可展示运行记录和结果。

边界：完成状态不等于回答全部正确；原序列错误仍需同时说明。

图 07 | 本地文献检索的真实界面

备份素材 | 历史本地检索界面，不当作最新完整问答



这是什么

本地小型文献库检索后显示来源片段的截图。

要说明什么

用户能否检索资料，并看到命中的来源内容？

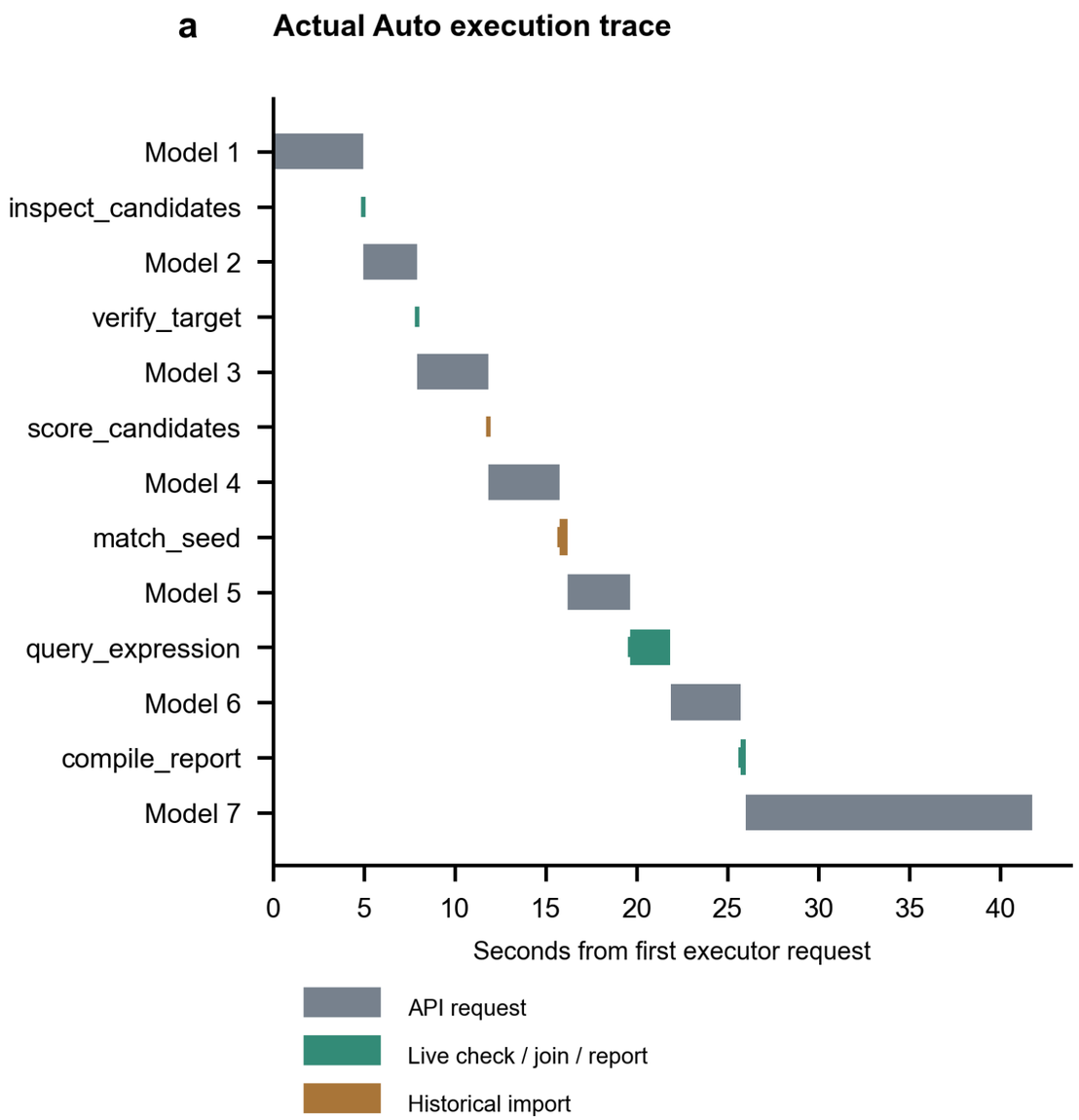
得到什么结论

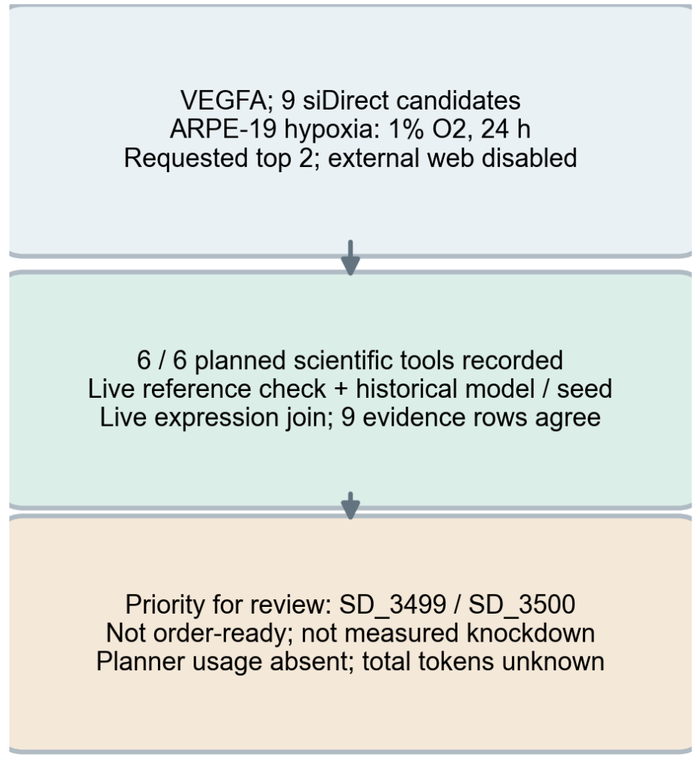
本地检索和来源片段展示已有实际入口。

边界：此截图不能单独证明模型读过来源；真实模型问答证据请用图02。

图 08 | 旧版真实 Auto 案例：仅作历史备份

历史素材 | 2026-10-20；原九条候选，区别于本次既购案例





Archived run 20261020_064921_2dae59; runtime e53bacb. Executor returned subtotal = 59,767 tokens; planning latency / usage not reconstructed.

这是什么

旧版Agent执行记录与原九条候选结果的组合图。

要说明什么

项目早期是否已有完整运行？前后版本如何区分？

得到什么结论

旧版曾完成六工具案例，但使用的是另一批输入。

边界：旧候选优先级不代表最新65条最佳；当时完整规划用量缺失，不能拿来比成本。

One purchased si-VEGF duplex, one recorded Agent run

Saved-record excerpts, not a web screenshot | run 20261024_121627_1e50c9 | code 03092ef

a User request and confirmed plan

Review the purchased duplex, preserve TT/N3, use the bound offline results,
check the reference, add expression context and produce a report. [Translated request excerpt]

Confirmed plan: candidate check; reference check; score; seed; expression; report.

8 model requests = 1 planning + 7 execution requests; 6 tool calls; 39,048 tokens.

b Execution log, 12:16:27-12:17:03 UTC

Call	Tool name (verbatim)	Recorded operation	Tool s
M1	inspect_candidates	Live input check	0.006
M2	verify_target	Live reference match	0.003
M3	score_candidates	Verified local import	0.002
M4	match_seed	Verified local import	0.536
M5	query_expression	Live baseline join	1.200
M6	compile_report	Report saved	0.127

M7: final model answer, 21.870 s; no seventh tool.

OligoFormer/SeedMatchR were imported, not rerun in the cloud.

c Evidence-derived candidate excerpt

Guide core, 19 nt (5'-3')

UUUCUUGUCUUGCUCUAUC

Other core, 19 nt (5'-3', conditional role)

GAUAGAGCAAGACAAGAAA

NM_003376.6: 1439-1457; coverage 18/20

OligoFormer score 1.121934 (not %)

Guide: 3,539 non-target sites / 2,839 genes

1,595 detected in hypoxia; 3 unmapped

Conditional roles; TT/N3 retained, not simulated.

Original answer error retained: the passenger core was written as 24 nt, not 19 nt.

Plan-to-execution gap 103.357 s (includes user wait; click time not logged); execution 36.040 s.

Planner duration not logged. Seed matches and expression are annotations, not measured siRNA effects.

Sources: 2026-10-24_nearfull_complete/live/; full original paths, versions and excerpts in SOURCE_MAP.md.

这是什么

Codex交付的未精简版既购案例证据图。

要说明什么

精简图中的细节能否回到完整记录?

得到什么结论

版本、时间、结果与错误均有完整说明。

边界：与图01是同一个案例，不作为新增样本或新模型回答。

图 10 | 原版完整文献图 B

追溯备份 | 保留完整文章句子、源码与译文引用

An answer linked to retrieved evidence

Recorded B02 query | 1 model call | 6 retrieved snippets | 6,232 tokens | code 179a7af

a User question (translated)

Can an OligoFormer score above 1 mean more than 100% knockdown?

b Evidence actually passed to the model

[S3] Article full text, text-c020

"The output of the model represents a prediction score for each siRNA, reflecting its potential effectiveness in inducing gene knockdown."

OligoFormer, DOI: 10.1093/bioinformatics/btae577

[S2] Official source code, text-c007

```
Y_PRED = [i*1.341 for i in Y_PRED]
RESULT['efficacy'] = Y_PRED
```

scripts/infer.py @ e2f53ad63387 (also retrieved in S5)

[S1] Project setting (translated, not a paper result)

"The model score is not an experimental knockdown percentage."

c Answer (translated excerpts)

No.

A score above 1 cannot mean more than 100% knockdown. [S1, S3]

The output is linearly scaled by 1.341. [S2, S5]

These snippets do not establish a measured knockdown percentage for a specific score.

English translations of saved Chinese records; original source quotes retained. Not a web screenshot.
Text fragments, not PDF pages. Source types remain separate. No independent quality score is claimed.
Run library_demo_20261024_032210_21c28f; library b4d516450903...; exact files and S1-S6 in SOURCE_MAP.md.

这是什么

Codex交付的未精简版B02证据图。

要说明什么

图02省略的文字与来源说明能否查到?

得到什么结论

全文摘录、代码、项目说明和回答引用均可追溯。

边界：与图02是同一道题，不增加独立题数，也不是专家评分。

这些图能支撑的论文表述

结论按证据层级表达，保留尚未完成的关键环节

图号	现在可以说	目前证据不支持
图01 / 图06	已完成一个真实网页Agent案例，能按确认计划连接六项工具。	系统总体可靠、回答完全正确、优于固定流程。
图02 / 图07	已有小文献库检索和一次有来源的真实模型问答。	文库覆盖全部领域文献，或模型总体准确率已得到验证。
图03	65条设计与单列既购序列的模型及seed结果已完整整理。	高分即高敲低率，或全部候选都适合订购/实验。
图04	264项查询全覆盖，23条设计存在指定条件的非靶近似匹配。	23条已证实脱靶，其余42条已获安全认证。
图05	自有未干预RNA-seq提供了本批次VEGFA表达背景。	已经验证siRNA敲低效果，或实现状态风险重排序。

下一项关键研究证据仍是独立人员对三组真实回答评分。当前图册没有代填评分，没有新增湿实验，没有调整候选规则或排序。

来源：main 7de40bd；完整路径与文件指纹见配套 SOURCE_MAP.md / source_manifest.json。